Rafael Antonio Tanda Pretel

Software Engineer — Backend-focused full-stack: APIs, frontend & databases

Havana, Cuba · relocating to La Palma, Spain · Spanish citizen · EU work authorization
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PROFILE

Software engineer, mostly backend, who also builds the frontend and database of each product, from requirements to tests, packaging and support. APIs and real-time services in Go and TypeScript; React PWA, Flutter, Godot and Python clients, offline-first for low-end hardware; data in PostgreSQL, Redis and SQLite. I work AI-first: I direct coding agents through written specs, reviewable units and tests and own every line that ships.

FREELANCE EXPERIENCE

Salon Manager — offline-first mobile app · Freelance · hair salon — Sep 2026

`Flutter, Dart, sqflite, Provider · github.com/Ranta2025/gestion-salon`

- Gathered requirements with the owner and delivered finances, clients, appointments with reminders, cash closing and PDF/CSV reports, fully on-device with no backend or recurring cost.
- Repositories tested against real in-memory SQLite; restore validates the backup before touching data.

Salary Fund Distribution — Acuerdo 10199/2025 · Freelance · client organization — Oct 2026

`Python, tkinter, pytest, PyInstaller · github.com/Ranta2025/fondo-salario-acuerdo-10199`

- Turned a 2025 Cuban regulation into an offline desktop app that distributes unexecuted salary funds, with configurable weights so each unit can match its own rules.
- Totals reconcile to the cent (largest-remainder rounding); domain logic isolated and covered by 100+ pytest tests; auditable Excel export.

AI-ASSISTED ENGINEERING

- Agent-driven workflow: requirements become numbered specs; coding agents (Claude Code, opencode) implement them in reviewable units; every change passes tests, multi-lens AI review and my own diff review.
- Built agent-bridge (github.com/Ranta2025/agent-bridge, MIT): links Claude Code to gentle-shell agents (orchestrated by gentle-ai) and tracks tasks to completion — state machine with delivery confirmation and retries, policy-driven auto-approver that never allows push, force or secret paths, RAM guard, 197 tests.
- Extended the tools: a Claude Code side-panel mod, an opencode background-agents plugin and custom skills.

SELECTED PROJECTS

ChatApp — real-time messaging platform — Jan 2026 – present

`Go, Gin, React 19, PostgreSQL, Redis, MinIO, WebSockets, Docker, Playwright · github.com/Ranta2025/chatapp`

- Built the Go API, React PWA and PostgreSQL/Redis data of a 7-service platform: 1:1 and group chat, receipts, presence, media, voice notes, video calls and Web Push; group permissions enforced server-side.
- Hardened auth (15-min JWTs in HttpOnly cookies, rotated refresh tokens, lockout, rate limits); PWA offline outbox delivers queued messages in order; 276 test files plus Playwright e2e in CI.

OrtoSim CUBA — offline medical training game (undergraduate thesis) — Oct 2026 – present

`Godot 4, GDScript, GUT, PocketBase, Caddy, JSON Schema, Docker`

- Serious game for 2 GB RAM Android phones under hard budgets (< 60 MB APK, 30 FPS, < 100 draw calls); hexagonal domain behind five ports, 150 GUT test files.
- Data-driven case engine (JSON Schema, 12 step kinds) shipped as signed content packs with a SHA-256 manifest; offline event queue synced via PocketBase.

SKILLS

Languages	Go, TypeScript, Python, Dart, GDScript, Java
Backend	Gin, REST, WebSockets, JWT auth, PocketBase
Frontend	React 19, PWA (service workers, IndexedDB, Web Push), Flutter, Godot 4
Databases	PostgreSQL, Redis, SQLite, IndexedDB, S3/MinIO, schemas & migrations
AI tooling	Claude Code, opencode, MCP, agent skills/hooks/plugins, spec-driven dev, AI review
Quality & delivery	go test, Playwright, pytest, GUT, GitHub Actions, Docker Compose, Nginx, Caddy

EDUCATION · LANGUAGES

Education	B.Eng. in Computer Science, University of Information Sciences (UCI), Havana — 3rd of 4 years, expected 2028
Languages	Spanish — native · English — B1 (certification exam March 2027)